

Indoor three-dimensional location estimation based on LED visible light communication

S.-H. Yang, E.-M. Jeong, D.-R. Kim, H.-S. Kim, Y.-H. Son and S.-K. Han

A novel concept for integrating visible light communications (VLC) with three-dimensional indoor positioning is presented. A VLC link based on transmitter and receiver characteristics using experimental measurements was modelled. Proposed is a three-dimensional positioning algorithm using received signal strength indication, which changes based on the angle and distance of the location based service. To reduce inter-cell interference, the transmitter's location code was sent using different subcarriers. A demonstration shows that the proposed algorithm can obtain a user's position, including height, accurately and without inter-cell interference.

Introduction: White light emitting diodes (LEDs) have been widely used in various fields such as illumination, communications and sensing technology. Visible light communication (VLC) is considered an attractive technology because it can be used for both lighting and communication at the same time [1, 2]. VLC has been verified as being suitable for various applications such as wireless communications, navigation, intelligent transportation systems and broadcasting.

Among the various services based on VLC, indoor positioning systems have many advantages: high security, no electromagnetic interference and high capacity compared to existing technologies using RF, ultrasound and infrared signals [3]. Also, positioning techniques using an image sensor based on visible light is very complicated and its implementation cost is very expensive [4].

In this Letter, a three-dimensional positioning system based on visible light is proposed. Each LED uses a different RF subcarrier to reduce a possible inter-cell interference by adjacent LEDs. The modelling of the link schematic characteristic was performed based on experimental measurements. A positioning error of less than 3 cm was demonstrated in a 2 m high 1.5 × 1.5 m cell space.

2D positioning algorithm: In general, the relationship between transmission distance and the RF power of the received optical signal can be expressed as [5]

$$P_r = (P_t/d^2) R_0(\theta) A_{eff}(\psi) \quad (1)$$

where P_r is the received optical power in a line-of-sight environment, P_t is average transmitted optical power, and d is the distance between the transmitter and the receiver. $R_0(\theta)$ is transmitted radiant intensity by radiation angle θ , and $A_{eff}(\psi)$ is the effective received area by incidence angle ψ .

The photocurrent i_{PD} at the photodetector is proportional to the received optical power. The RF power received from the transmitter is [6]

$$P_{RF} = (C_{RF}/d^4) P_t^2 \quad (2)$$

where C_{RF} is the RF power constant, including the optical to electrical conversion efficiency and the optical power constant. The proportional relation among the photocurrent, the received optical power, and the RF power can thus be expressed as

$$i_{PD}^2 \propto P_r^2 \propto P_{RF} \quad (3)$$

Received P_{RF} is determined by the transmission distance d and the RF power constant C_{RF} including angle gain. A user's location can thus be obtained using a trilateration method [7].

3D positioning algorithm: Angle-gain can be varied based on the lens shape of the transmitter and receiver. In the preceding Section, one-to-one correspondence between the angle and the transmission distance is satisfied because the vertical distance is fixed. However, in a three-dimensional positioning system, the angle can be changed while maintaining the same transmission distance. In this case, the transmission distance is the same even though the incidence and radiation angle are different. As a result, numerous vertical distances and horizontal distances are possible for the same P_{RF} . We calculated transmission distances through a compensation process using an angle-gain profile in

which the angle ranges from 0° to 90°. The incidence and radiation angle are the same when the transmitting and receiving planes are parallel. The modelled angle-gain curve according to incidence and radiation angle and based on experimental measurement is shown in Fig. 1.

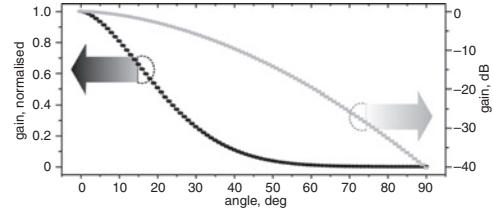


Fig. 1 Angle-gain profile of proposed system based on experimental measurement

Fig. 2 shows an indoor positioning system. The vertical distance and horizontal distance can be obtained by using transmission distance and angle according to the received P_{RF} and angle-gain profile. Thus, the vertical distance and the horizontal distance may not be unique. If that is the case, location estimation in three-dimensional space uses the following process. First, if we suppose that vertical distance from the transmitters is the same, we can obtain circles whose radius is the horizontal distance and whose centre is the position of each transmitter. Secondly, we compute the total area within overlapping circles. Finally, the vertical distance is searched when the extent of the possible area reaches a minimum. A larger possible area implies a less accurate estimated position. As a result, user position is estimated by both vertical distance and the numerical mean of cross points of circles.

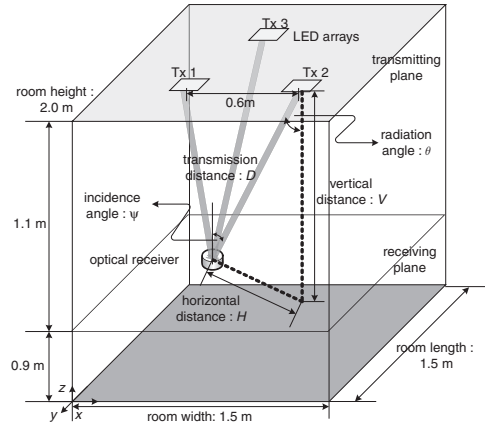


Fig. 2 Proposed three-dimensional positioning system based on LED arrays using different subcarriers to remove inter-cell interference

System modelling: As can be seen in Fig. 2, three LED arrays were used to illuminate a room; each LED transmits its coordinates simultaneously using 2.0, 2.5, and 3.0 MHz subcarriers to reduce possible inter-cell interference [6]. The room had dimensions of 1.5 by 1.5 m and a height of 2.0 m. The photodetector is positioned over 0.9 m from the floor. The distance between LEDs was 0.6 m. In this case, the positioning parameters can be expressed as

$$V_i = D_i \cos(\psi_i), \quad H_i = D_i \sin(\psi_i) \quad (4)$$

thus

$$H_i = V_i \tan(\psi_i) \quad (5)$$

where V_i is vertical distance, D_i is transmission distance and H_i is horizontal distance between Tx_i and Rx . ψ_i and θ_i are incidence angle and radiation angle, respectively. The incidence angle and the radiation angle have the same value when the transmitting plane and the receiving plane are parallel. When the received P_{RF} is compensated for angle-gain at a specific angle, D_i is obtained from (1) and (2). Vertical distance and horizontal distance are obtained from D_i and the specific angle. Thus, possible area varies as the angle changes.

Results: Angle and horizontal distance can be determined given the vertical distance at a specific received P_{RF} . Fig. 3a shows the change in the horizontal distance from each transmitter when vertical distance changes. Ideally, the intersection point should be a single dot, but in a real situation, quantisation and measurement error components are added. We need to find a specific vertical distance when the possible area, which is determined by the intersecting points, is at a minimum value. That can be expressed as

$$\min_V \left(\int_{I_{V23}}^{I_{V12}} (C_{V2} - C_{V1}) dx + \int_{I_{V31}}^{I_{V23}} (C_{V2} - C_{V3}) dx \right) \quad (6)$$

where I_{Vij} are the x -coordinates of i and j transmitter intersection points of circles. C_{Vi} is the circle whose centre is the coordinates of Tx_i and the radius is the horizontal distance at a given vertical distance V . Thus, C_{Vi} can be expressed as

$$C_{vi} = (x - x_i)^2 + (y - y_i)^2 - (H_{vi})^2 \quad (7)$$

where x_i and y_i are x and y coordinates of the i th Tx and H_{vi} is the horizontal distance at a given vertical distance and Tx_i . The centre of mass of the triangle whose vertices are three intersecting points can be estimated as the user's position. It is obtained by using the numerical mean of the triangle. For example, Fig. 3b shows that intersection points were (20.03, 30.29, 77.87), (19.24, 29.21, 77.21) and (23.55, 27.64, 77.44). The coordinates estimated using numerical means were (20.94, 29.05, 77.51) when the real coordinates were (20, 30, 80), which results in less than 3 cm of error in our positioning process. The system performance could be improved by reducing the quantisation error based on a delicately designed vertical distance with a finer granularity.

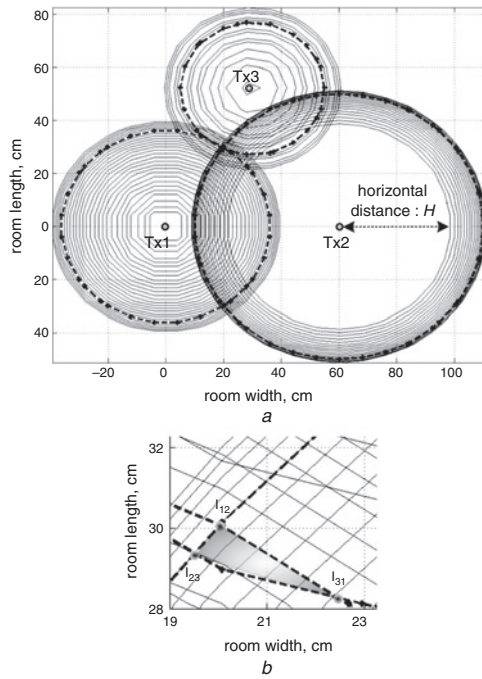


Fig. 3 Received RF power of Tx_1 , Tx_2 , and Tx_3 are -42.70 , -46.67 , and -39.79 dBm, respectively

a Change in horizontal distance with vertical distance
b Process of positioning using intersection points and surface area

Conclusions: We have proposed a simple and practical three-dimensional indoor positioning system which makes use of LEDs, which allows for both lighting and communications. The system is based on experimentally measured RSS with less than 3 cm of error. The proposed positioning method, which uses received RF power, could be an efficient solution for indoor location-based services by adding tilting and an angle-gain compensation process.

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